Light

GCSE Computer Science — Memory and Storage (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Secondary storage is an essential component of a computer system. Which statement best describes the purpose of secondary storage?

- ○ **A.** To store data permanently so it is not lost when the computer is turned off
- ○ **B.** To run programs faster than the CPU can process them
- ○ **C.** To perform calculations when the ALU is busy
- ○ **D.** To display output on the monitor

Q2 (1 mark)

A student buys a USB memory stick to transfer files between home and school.
What type of storage technology does a USB memory stick use?

- ○ **A.** Magnetic storage
- ○ **B.** Optical storage
- ○ **C.** Volatile storage
- ○ **D.** Solid state storage

Q3 (1 mark)

A desktop computer contains a hard disk drive (HDD) for storing files.
What type of storage technology does an HDD use?

- ○ **A.** Optical storage
- ○ **B.** Solid state storage
- ○ **C.** Cloud storage
- ○ **D.** Magnetic storage

Q4 (1 mark)

A film is distributed on a Blu-ray disc.
What type of storage technology does a Blu-ray disc use?

- ○ **A.** Magnetic storage
- ○ **B.** Solid state storage
- ○ **C.** Optical storage
- ○ **D.** Flash storage

Q5 (1 mark)

All computer data is stored as binary digits.
What is the term for a single binary digit that can be either 0 or 1?

- ○ **A.** A bit
- ○ **B.** A byte
- ○ **C.** A nibble
- ○ **D.** A pixel

Q6 (1 mark)

File sizes are usually measured in bytes rather than bits.
How many bits make up one byte?

- ○ **A.** 8 bits
- ○ **B.** 4 bits
- ○ **C.** 16 bits
- ○ **D.** 1024 bits

Q7 (1 mark)

A photographer wants to share images online but the files are very large.
What process can reduce the file size to make sharing easier?

- ○ **A.** Compression
- ○ **B.** Encryption
- ○ **C.** Fragmentation

○ **D.** Installation

Q8 (1 mark)

A company backs up important spreadsheet files using compression. Why must they use lossless compression rather than lossy compression?

○ **A.** Lossless compression creates smaller files than lossy compression

○ **B.** Lossless compression works faster than lossy compression

○ **C.** Lossless compression is the only type that works on spreadsheets

○ **D.** Lossless compression allows the exact original data to be restored

Q9 (1 mark)

A streaming music service uses MP3 format for its audio files. MP3 is a lossy compression format. What does this mean?

○ **A.** The files can only be played once before they are deleted

○ **B.** The compression process is unreliable and often fails

○ **C.** The files become larger after compression

○ **D.** Some audio data is permanently removed to reduce file size

Q10 (1 mark)

An SSD has no moving parts inside it.
What advantage does this provide compared to a hard disk drive?

- ○ **A.** SSDs can store more data in the same physical space
- ○ **B.** SSDs are less likely to be damaged by dropping or shaking the computer
- ○ **C.** SSDs are cheaper to manufacture than hard disk drives
- ○ **D.** SSDs can connect to more types of computers

Q11 (1 mark)

A business needs to store a large amount of data as cheaply as possible. Which type of storage typically offers the lowest cost per gigabyte?

- ○ **A.** Solid state drives (SSDs)
- ○ **B.** USB flash drives
- ○ **C.** Magnetic hard disk drives (HDDs)
- ○ **D.** Blu-ray discs

Q12 (1 mark)

Software companies sometimes distribute programs on CD-ROM or DVD-ROM.
What does ROM mean in this context?

- ○ **A.** Randomly Operated Memory
- ○ **B.** Read Only Memory - the disc cannot be written to by the user
- ○ **C.** Repeatedly Overwritable Media
- ○ **D.** Rapidly Operating Machine

Q13 (2 marks)

A hard drive has a capacity of 2 tebibytes (TiB).
How many gibibytes (GiB) is this equivalent to?

- ○ **A.** 2000 GiB, because 1 TiB = 1000 GiB
- ○ **B.** 0.00195 GiB, because you divide by 1024
- ○ **C.** 2,097,152 GiB, because you multiply by 1024 twice
- ○ **D.** 2048 GiB, because 1 TiB = 1024 GiB

Q14 (2 marks)

A file is 5120 kibibytes (KiB) in size.
How many mebibytes (MiB) is this?

- ○ **A.** 5.12 MiB, because 5120 ÷ 1000 = 5.12
- ○ **B.** 0.005 MiB, because KiB is much smaller than MiB
- ○ **C.** 5,242,880 MiB, because 5120 × 1024 = 5,242,880
- ○ **D.** 5 MiB, because 5120 ÷ 1024 = 5

Q15 (2 marks)

A computer technician is explaining the differences between SSDs and HDDs to a customer.
Which statement correctly compares these storage types?

- ○ **A.** HDDs access data faster than SSDs because spinning disks can locate data more quickly
- ○ **B.** SSDs are slower but more reliable because they use older,

proven technology

○ **C.** HDDs and SSDs have identical access speeds because they use the same interface

○ **D.** SSDs access data faster than HDDs because they have no mechanical parts to wait for

Q16 (2 marks)

A software developer needs to choose between CD, DVD, and Blu-ray for distributing a 6 GiB program.
Which optical format has sufficient capacity?

○ **A.** CD, because a standard CD holds approximately 700 MiB

○ **B.** DVD, because a standard single-layer DVD holds approximately 4.7 GiB

○ **C.** Blu-ray, because a standard single-layer Blu-ray holds approximately 25 GiB

○ **D.** Any of them, because optical discs can expand to fit the data

Q17 (2 marks)

A game developer needs to distribute a game that includes both program files and background music.
Which compression approach is most appropriate?

○ **A.** Use lossy compression for everything because it creates the smallest files

○ **B.** Use lossless compression for everything because quality is always the priority

○ **C.** Use lossless compression for program files (to preserve exact data) and lossy compression for music (where some quality loss is

acceptable)

○ **D.** Use no compression because modern storage is large enough for uncompressed files

Q18 (2 marks)

When a hard disk drive reads data, a mechanical component moves across the spinning platters.
What is this component called, and what does it do?

○ **A.** The spindle motor, which spins the read/write head to generate data

○ **B.** The read/write head, which detects or changes the magnetic orientation of areas on the platter

○ **C.** The laser assembly, which burns data onto the magnetic surface

○ **D.** The flash controller, which converts magnetic signals to electrical charges

Q19 (2 marks)

A user wants to burn their own music compilation onto a disc that they can add more songs to later.
Which type of optical disc should they use?

○ **A.** CD-ROM, because it has the highest audio quality

○ **B.** CD-R, because it can be written to once and is compatible with most players

○ **C.** CD-RW, because it can be written to multiple times and erased

○ **D.** DVD-ROM, because it has more storage space for music

Q20 (2 marks)

A folder contains 8 photos, each exactly 1.5 MiB in size.
What is the total size of all photos in the folder?

- ○ **A.** 9.5 MiB, because 8 + 1.5 = 9.5
- ○ **B.** 12 MiB, because 8 × 1.5 = 12
- ○ **C.** 5.33 MiB, because 8 ÷ 1.5 = 5.33
- ○ **D.** 12,288 KiB, because 8 × 1.5 = 12 MiB = 12,000 KiB

Q21 (2 marks)

A computer uses RAM for running programs and an SSD for storing files.
Which statement correctly compares these two types of memory?

- ○ **A.** RAM is non-volatile and permanent; SSD is volatile and temporary
- ○ **B.** RAM is faster but volatile; SSD is slower but non-volatile
- ○ **C.** RAM and SSD are identical in speed and volatility
- ○ **D.** RAM stores more data than SSD because it is closer to the CPU

Q22 (2 marks)

An original file is 10 MiB. After lossless compression, it is 2 MiB.
What is the compression ratio?

- ○ **A.** 2:10, meaning the compressed file is 2 parts for every 10 parts original

○ **B.** 5:1, meaning the original is 5 times larger than the compressed file

○ **C.** 1:5, meaning each original byte becomes 5 compressed bytes

○ **D.** 8:2, meaning 8 MiB was removed to leave 2 MiB

Q23 (2 marks)

A database administrator is concerned about using SSDs for a database with millions of write operations per day.
Why might this be a concern?

○ **A.** SSDs cannot handle more than 100 write operations per day

○ **B.** Flash memory cells can only be written to a limited number of times before wearing out

○ **C.** SSDs are read-only devices that cannot accept write operations

○ **D.** Writing to SSDs is slower than writing to HDDs

Q24 (3 marks)

A digital camera takes photos with these settings:
• Resolution: 4000 × 3000 pixels
• Colour depth: 24 bits per pixel
What is the uncompressed file size of one photo in bytes?

○ **A.** 36,000,000 bytes, because 4000 × 3000 × 3 = 36,000,000

○ **B.** 12,000,000 bytes, because 4000 × 3000 = 12,000,000 pixels

○ **C.** 288,000,000 bytes, because 4000 × 3000 × 24 = 288,000,000

○ **D.** 1,000,000 bytes, because 4000 × 3000 ÷ 12 = 1,000,000

Q25 (3 marks)

A server has 4 TiB of storage capacity.
How many bytes is this?

○ **A.** 4,000,000,000,000 bytes, because 1 TiB = 1,000,000,000,000 bytes

○ **B.** 4,398,046,511,104 bytes, because 1 TiB = 1024^4 bytes

○ **C.** 4,294,967,296 bytes, because 1 TiB = 1024^3 bytes

○ **D.** 4,194,304 bytes, because 1 TiB = 1024^2 bytes

Q26 (3 marks)

An audio recording has these properties:
• Sample rate: 48,000 samples per second
• Bit depth: 16 bits per sample
• Duration: 2 minutes
• Stereo (2 channels)
What is the uncompressed file size in bytes?

○ **A.** 23,040,000 bytes, because 48,000 × 16 × 120 × 2 ÷ 8 = 23,040,000

○ **B.** 11,520,000 bytes, because 48,000 × 16 × 120 ÷ 8 = 11,520,000 (ignoring stereo)

○ **C.** 184,320,000 bytes, because 48,000 × 16 × 120 × 2 = 184,320,000 (without bit conversion)

○ **D.** 1,536,000 bytes, because 48,000 × 16 × 2 ÷ 8 = 1,536,000 (per second only)

Q27 (3 marks)

A video file is 800 MiB uncompressed. After lossy compression, it is 50 MiB.
What is the compression ratio and how much space is saved?

- ○ **A.** Compression ratio 16:1; space saved 750 MiB (93.75%)
- ○ **B.** Compression ratio 1:16; space saved 50 MiB (6.25%)
- ○ **C.** Compression ratio 50:800; space saved 800 MiB (100%)
- ○ **D.** Compression ratio 8:1; space saved 700 MiB (87.5%)

Q28 (3 marks)

A hospital needs to select storage for patient scan images.
Requirements:
• Images must never be lost or corrupted
• Fast access for emergencies
• Storage for 500 GiB of data
• Budget is not the primary concern
Which storage solution best meets these requirements?

- ○ **A.** Multiple DVD-RWs, because optical media is the most reliable
- ○ **B.** A single large HDD, because magnetic storage offers the best balance of speed and reliability
- ○ **C.** A high-quality SSD with regular backups to a second SSD, because SSDs provide fast access and redundant backup prevents data loss
- ○ **D.** USB flash drives, because they are portable and can be stored in different locations

Q29 (3 marks)

A backup system needs to store 640 GiB of data. The available drive has 655,360 MiB of free space.
Will all the data fit on the drive?

- ○ **A.** No, because 655,360 MiB is less than 640 GiB when converted correctly
- ○ **B.** Yes, because 655,360 MiB = 640 GiB exactly, so all data fits
- ○ **C.** No, because MiB and GiB use different number systems and cannot be compared
- ○ **D.** Yes, because 655,360 is a larger number than 640

Q30 (3 marks)

A company needs storage solutions for three different purposes:
• Purpose A: Long-term archive of old financial records (rarely accessed)
• Purpose B: Operating system and applications (fast boot and load times needed)
• Purpose C: Distributing training videos to branch offices
Which combination of storage technologies is most appropriate?

- ○ **A.** A: SSD; B: HDD; C: USB flash drives
- ○ **B.** A: HDD; B: SSD; C: DVD or Blu-ray
- ○ **C.** A: DVD-ROM; B: DVD-RW; C: HDD
- ○ **D.** A: USB flash drives; B: CD-ROM; C: SSD

Total: **55** marks • Time: **30 min**

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